

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$M dx + N dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int M dx + \int (\text{terms in } N \text{ free from } x) dy = C$$

Let $D\text{-}\mathcal{E}_n$ be $M dx + N dy = 0$

$$\& \text{ if } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\Rightarrow D\text{-}\mathcal{E}_n$ is nonexact.

Exact $D\text{-}\mathcal{E}_n = (D\text{-}\mathcal{E}_n \text{ Nonexact}) \times \text{Integrating factor. (IF)}$

① If $M dx + N dy = 0$ is homog. $D\text{-}\mathcal{E}_n$

(i.e. M & N homog. of same deg.)

$$\text{then I.f.} = \frac{1}{Mx + Ny} \quad Mx + Ny \neq 0$$

② If $D\text{-}\mathcal{E}_n M dx + N dy = 0$ is in the form.

$$y f_1(xy) dx + x f_2(xy) dy = 0$$

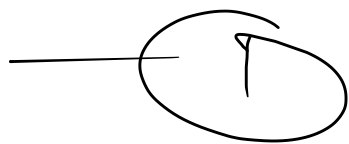
$$\text{then I.f.} = \frac{1}{Mx - Ny} \quad Mx - Ny \neq 0$$

$$\begin{array}{c} t \\ \uparrow \\ f(x, y) \\ x^2 + y^2 \\ x^2 y^2 + 2xy + 1 \end{array}$$

$$\textcircled{3} \quad \text{If } \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x) \quad \text{then I.f.} = e^{\int f(x) dx}$$

$$\textcircled{4} \quad \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y) \quad \text{then I.f.} = e^{\int f(y) dy}$$

Solve $y(1+xy)dx + x(1-xy)dy = 0$



$$M = y + xy^2 \quad N = x - x^2y$$

$$\frac{\partial M}{\partial y} = 1 + 2xy \quad \frac{\partial N}{\partial x} = 1 - 2xy$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$\therefore$ D.Eqⁿ is non exact.

Given D.Eqⁿ is in the form

$$y f_1(xy) dx + x f_2(xy) dy = 0$$

$$I.f = \frac{1}{Mx - Ny} = \frac{1}{xy + x^2y^2 - (xy - x^2y^2)} = \frac{1}{2x^2y^2}$$

Multiplying D.Eqⁿ by I.f.

$$\frac{1}{2x^2y^2} y(1+xy) dx + \frac{1}{2x^2y^2} x(1-xy) dy = 0$$

$$\left(\frac{1}{2x^2y} + \frac{1}{2x} \right) dx + \left(\frac{1}{2xy^2} - \frac{1}{2y} \right) dy = 0$$

D.Eqⁿ is exact

$$M' = \frac{1}{2x^2y} + \frac{1}{2x} \quad N' = \frac{1}{2xy^2} - \frac{1}{2y}$$

$$\int M' dx + \int (\text{terms in } N' \text{ free from } x) dy = C$$

$$\int \left(\frac{1}{2x^2y} + \frac{1}{2x} \right) dx + \int -\frac{1}{2y} dy = C$$

$$\int x^{-2}$$

$$\frac{1}{2y} \frac{x^{-1}}{(-1)} + \frac{1}{2} \log x - \frac{1}{2} \log y = C$$

$$-\frac{1}{2xy} + \log x - \log y = 2C$$

$$-\frac{1}{2xy} + \log\left(\frac{x}{y}\right) = C \quad \checkmark$$

Solve $(4xy + 3y^2 - x)dx + x(x + 2y)dy = 0$

$$M = 4xy + 3y^2 - x$$

$$N = x^2 + 2xy$$

$$\frac{\partial M}{\partial y} = 4x + 6y \neq \frac{\partial N}{\partial x} = 2x + 2y$$

non exact.

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4x + 6y - (2x + 2y)}{x(x + 2y)} = \frac{2x + 4y}{x(x + 2y)}$$

$$= \frac{2(x + 2y)}{x(x + 2y)} = f(x)$$

$$I.F. = e^{\int f(x) dx} = e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$$

multiplying by I.F.

$$x^2(4xy + 3y^2 - x)dx + x^3(x + 2y)dy = 0$$

$$M' = 4x^3y + 3x^2y^2 - x^3 \quad N' = x^4 + 2x^3y$$

clearly it is exact D.E.

Soⁿ.

$$\int (M' \text{ free from } y) dx + \int N' dy = C$$

$$\Rightarrow \int -x^3 dx + \int (x^4 + 2x^3y) dy = C$$

$$\Rightarrow -\frac{x^4}{4} + x^4y + \frac{2x^3y^2}{2} = C$$

Solve $(3xy^2 - y^3)dx - (2x^2y - xy^2)dy = 0$

$$M = 3xy^2 - y^3 \quad N = -(2x^2y - xy^2)$$

$$\frac{\partial M}{\partial y} = 6xy - 3y^2 \neq \frac{\partial N}{\partial x} = -(4xy - y^2)$$

Not exact.

M & N homog. of deg 3

$$\therefore IF = \frac{1}{Mx + Ny}$$

$$= \frac{1}{3x^2y^2 - \cancel{xy^3} - 2x^2y^2 + \cancel{xy^3}}$$

$$= \frac{1}{x^2y^2}$$

Multiplying by an IF

$$\frac{(3xy^2 - y^3)}{x^2y^2} dx - \frac{(2x^2y - xy^2)}{x^2y^2} dy = 0$$

Clearly Exact

$$M' = \frac{3}{x} - \frac{y}{x^2} \quad N' = -\frac{2}{y} + \frac{1}{x}$$

So it is

$$\int M' dx + \int (\text{term in } N' \text{ free from } x) dy = C$$

$$\Rightarrow \int \left(\frac{3}{x} - \frac{y}{x^2}\right) dx + \int -\frac{2}{y} dy = C$$

x^{-2}

$$\Rightarrow 3 \log x + \frac{y}{x} - 2 \log y = C$$

$\downarrow M$
 $\uparrow N$

Solve $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$

$$\frac{\partial M}{\partial y} = 12x^2y^3 + 2x \neq \frac{\partial N}{\partial x} = 6x^2y^3 - 2x \quad \text{non exact}$$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{6x^2y^3 - 2x - 12x^2y^3 - 2x}{3x^2y^4 + 2xy}$$

$$= \frac{-6x^2y^3 - 4x}{xy(3xy^3 + 2)} = \frac{-2x(3xy^3 + 2)}{xy(3xy^3 + 2)}$$

$$I.f. = e^{\int f(y) dy} = e^{\int -2/y dy} = e^{-2 \log y} = y^{-2} = \frac{1}{y^2}$$

Multiplying by a I.f.

$$\left(\frac{3x^2y^4 + 2xy}{y^2} \right) dx + \left(\frac{2x^3y^3 - x^2}{y^2} \right) dy = 0$$

exact

$$M' = 3x^2y^2 + \frac{2x}{y}$$

$$N' = 2x^3y - \frac{x^2}{y^2}$$

Solⁿ is

$$\int M' dx + \int (\text{term in } N' \text{ free from } x) dy = C$$

$$\int \left(3x^2y^2 + \frac{2x}{y} \right) dx + 0 = C$$

$$\frac{3x^3y^2}{3} + \frac{2x^2}{y} = C$$

$$\left[y \left(1 + \frac{1}{x} \right) + \cos y \right] dx + (x + \log x - x \sin y) dy = 0$$

$$(x^4 e^x - 2mxy^2) dx + 2mx^2 y dy = 0$$

• Solve $\frac{dy}{dx} = -\frac{x^2 y^3 + 2y}{2x - 2x^3 y^2}$

$$(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$$

$$y(x + y) dx - x(y - x) dy = 0$$

$$[y(1 + \frac{1}{x}) + \cos y] dx + (x + \log x - x \sin y) dy = 0$$

$$\frac{\partial M}{\partial y} = 1 + \frac{1}{x} - \sin y$$

$$\frac{\partial N}{\partial x} = (1 + \frac{1}{x}) - \sin y$$

$$(x^4 e^x - 2mxy^2) dx + 2mx^2 y dy = 0$$

$$\frac{\partial M}{\partial y} = -2mx^2 y \neq \frac{\partial N}{\partial x} = 2my^2 x$$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{-4mxy - 4mxy}{2mxy^2}$$

$$= \frac{-8mxy}{2mxy^2}$$

$$= -\frac{4}{y} = f(y)$$

$$I.f = e^{-\int \frac{4}{y} dy}$$

$$= e^{-4 \log x}$$

$$= e^{\log x^{-4}} = \frac{1}{x^4}$$

$$y(x+y) dx - x(y-x) dy = 0$$

$$(xy + y^2) dx - (xy - x^2) dy = 0$$

$$M = xy + y^2 \quad N = -(xy - x^2)$$

$$\frac{\partial M}{\partial y} = x + 2y \neq \frac{\partial N}{\partial x} = -(y - 2x)$$

as D & N homog.

$$I.f = \frac{1}{Mx + Ny}$$

$$= \frac{1}{(xy + y^2)x - (xy - x^2)y}$$

$$= \frac{1}{x^2 y + xy^2 - xy^2 + x^2 y} = \frac{1}{2x^2 y}$$

Mult.

$$(y^4 + 2y) dx + (xy^3 + 2y^4 - 4x) dy = 0$$

$$\frac{\partial M}{\partial y} = 4y^3 + 2 \neq \frac{\partial N}{\partial x} = y^3 - 4$$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{4y^3 + 2 - y^3 + 4}{(xy^3 + 2y^4 - 4x)}$$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y^3 - 4 - 4y^3 - 2}{y^4 + y}$$

$$I.f = e^{\int -3y dy}$$

$$= e^{-3 \log y}$$

$$= e^{\log y^{-3}} = \frac{1}{y^3}$$

$$= \frac{-3y^3 - 6}{y(y^3 + 2)} = \frac{-3(y^3 + 2)}{y(y^3 + 2)} = f(y)$$

$$\bullet \text{ Solve } \frac{dy}{dx} = -\frac{x^2 y^3 + 2y}{2x - 2x^3 y^2}$$

$$(2x - 2x^3 y^2) dy = -(x^2 y^3 + 2y) dx$$

$$(x^2 y^3 + 2y) dx + (2x - 2x^3 y^2) dy = 0$$

$$\frac{\partial M}{\partial y} = x^2 3y^2 + 2$$

$$\frac{\partial N}{\partial x} = 2 - 2y^2 3x^2 = 2 - 6x^2 y^2$$

$$y(x^2 y^2 + 2) dx + x(2 - 2x^2 y^2) dy = 0$$

$$y f_1(xy) dx + x f_2(xy) dy = 0$$

$$I.f = \frac{1}{Mx - Ny} = \frac{1}{(x^2 y^3 + 2y)x - (2x - 2x^3 y^2)y}$$

$$= \frac{1}{x^3 y^3 + 2xy - 2xy + 2x^3 y^3}$$

$$I.f = \frac{1}{3x^3 y^3} //$$